

Microprocessors 1

January 23rd, 2026

Deadline: January 30th, 2026

Name: _____ DU ID: _____

LAB 3

Overview

The main purpose of this lab is to test concepts of interrupt routines, Timers, GPIO input/output and volatile variables.

Submission Directions

You need to submit your work using the GitHub link that you created for this class. Don't forget to include comments throughout your code.

Exercise

Build the program with the following specs:

1. You will have 2 tasks running simultaneously and 1 foreground task (the main task).
2. Each task has a respective timer running at a certain frequency (see table below).

	Frequency (Hz)	Timer
Task 1	1	1
Task 2	10	2

3. Task 1 is responsible for blinking the Green LED at a rate of 1 Hz. It is also responsible for incrementing a counter that will be used by the main task as a "timer".
4. Task 2 (running at 10 Hz) is responsible for reading an input pin (Button) and if it is equals to 1 it turns on the Red LED otherwise it turns off the Red LED.
5. The input pin can be either Button 1 or Button 2 on your board.
6. The main task (foreground task) prints over the serial port every 3 seconds a message of your choice. It uses the counter from task 1 as a "timer".
[Note: pay attention to variables that should be declared as volatile]
7. Provide a block diagram of your code showing how the tasks interact with each other.